Comparison of Eye-Tracking Representations for Engagement Inference in Office Workers

Tomi Božak
Jožef Stefan Institute
Ljubljana, Slovenia
tb85088@student.uni-lj.si

Gašper Slapničar
Jožef Stefan Institute
Ljubljana, Slovenia
gasper.slapnicar@ijs.si

Abstract

Eye tracking is an appealing sensor for inferring momentary user states at work. Before training a model, a practitioner must decide how to represent a short gaze window. We compare four representation families on identical three-second windows from an ongoing office-worker study: raw multichannel signals, handcrafted ocular features, frozen GazeMAE embeddings and frozen MOMENT embeddings. The target is self-reported momentary engagement. All four are evaluated under leave-one-subject-out cross-validation with fold-local preprocessing, using a random forest classifier and an MLP against a majority baseline. Raw signals give the best unseen-participant performance (0.538 accuracy, 0.014 above the baseline), handcrafted features follow closely, and neither frozen encoder improves on them. Absolute discrimination is weak throughout (best ROC-AUC 0.551) and between-participant variability dwarfs the differences between representations, so we report the ranking as preliminary, cost-aware guidance.

Keywords

eye tracking, engagement inference, representation learning, GazeMAE, MOMENT, subject generalisation

1 Introduction

Interactive systems that adapt to the person using them need evidence about that person's momentary state. In knowledge work, engagement (how absorbed someone is in the task at hand) is an attractive target. It fluctuates within a working day, it is only weakly visible in application logs, and it is the state most directly tied to productivity and well-being interventions. Eye tracking is a plausible sensor for it. Gaze position, fixation and saccade dynamics, blinks and pupil size are available continuously and non-intrusively from a screen-mounted tracker, require no action from the user, and have documented links to attention and processing load.

Turning a stream of gaze samples into model input forces a design decision that is rarely examined on its own (cite). A short window of eye-tracking data can be presented to a classifier in at least four ways. The raw multichannel signal retains everything the tracker measured, at the price of high dimensionality and no structure. Handcrafted ocular features, including fixation, saccade, blink and pupil statistics, encode decades of domain knowledge, stay interpretable, and remain the default choice in applied work [3, 2](cite also someone else beside bozak). Learned representations promise to remove the manual step. GazeMAE [1] provides gaze-specific embeddings pretrained on eye-movement corpora, while general-purpose time-series foundation models such as MOMENT [4] provide a frozen encoder intended to transfer to unseen domains. The four options differ substantially in preprocessing effort, storage and compute, yet they are seldom compared under the same experimental conditions. Published results differ in windows, labels, classifiers and, most consequentially, in validation protocol, so an apparent advantage of one representation can just as easily come from an optimistic split. (cite) We therefore ask which eye-tracking representation supports the most reliable inference of self-reported momentary engagement in unseen office workers. We answer with a controlled comparison of raw signals, handcrafted features, GazeMAE embeddings and MOMENT embeddings on identical three-second windows, an identical binary engagement target and identical leave-one-subject-out folds. Both pretrained encoders are kept frozen and only a prediction head is trained, which is the out-of-the-box setting a practitioner would try first. This design provides cost-aware guidance for representation selection under limited labelled data.

2 Related Work

Eye tracking has long been used as behavioural evidence for internal user states. Pupil dilation is a classical correlate of processing load [5], and fixation, saccade, blink and pupil statistics computed over short windows remain the dominant input to models of workload, affect and engagement.(cite) Our own prior work illustrates the appeal and the limits of this pipeline. Handcrafted ocular features supported coarse affective distinctions [3]. A systematic evaluation across several prediction targets also showed that task type is recognised far more reliably than graded workload, and that reported accuracy depends strongly on how the prediction target and the validation split are defined [2].

Learned representations offer an alternative. GazeMAE pretrains temporal autoencoders on raw position and velocity signals and shows that the resulting gaze-specific embeddings support downstream classification through a simple linear head [1]. MOMENT takes a broader approach by pretraining a multichannel encoder for transfer across time-series domains without task-specific pretraining [4]. The value of both encoders for modest, subject-dependent recordings of spontaneous behaviour remains unclear.

Evaluation design is a further concern. Windows drawn from the same person are highly dependent, so random window splits and preprocessing fitted on the full dataset inflate reported performance, an effect documented in detail for other biosignal domains [6]. Comparisons across representations are interpretable only when every candidate is evaluated on identical windows and labels under a subject-disjoint protocol with fold-local preprocessing.

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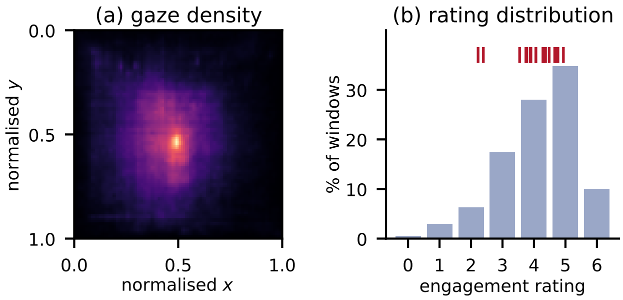

Figure 1: Descriptive context for the modelling cohort. (a) Gaze-coordinate density on the normalised display, pooled over all on-screen samples. (b) Distribution of the engagement rating over the 140,531 modelled windows. The red ticks mark the 17 individual participant means, which span 2.2–4.9 and motivate the within-participant centring of the target.

3 Methodology

3.1 Data and Eye-Tracking Representations

The data come from a multimodal study in which office workers are recorded during ordinary work at their own workplace. The wider collection comprises an RGB camera, a screen-mounted eye tracker, a microphone and screen-activity logs. This paper uses the eye-tracking modality only. Data collection is ongoing and the dataset is not yet public. This preliminary study uses the cohort available at the time of writing, which includes 17 participants and 657 full-workday recordings, with 19 to 79 recordings per participant (median 32). The tracker samples at 60 Hz, and we retain five channels: left and right pupil size, gaze x and y, and the average eye–screen distance.

Labels come from an experience-sampling protocol. Participants are prompted several times per working day with a short questionnaire covering engagement, affect, fatigue, stress, physical comfort, workplace interactions, sleep and productivity. Our target is the absorption item, “Right now I am immersed in my work”, one of three momentary engagement items answered at every prompt and rated on a seven-point scale from 0 to 6. We call it the *engagement rating*. Each prompt labels the 15 minutes of recording that precede it, and only those intervals enter this study.

(change the color of the heatmap to blue-red and for the consistency, use the same blue and red in the histogram if possible (not strictly mandatory))

Recordings are segmented into three-second windows with a 2.25-second step (25 % overlap). Every window carries an identifier built from the participant, the source recording and the position of the window inside that recording. The four representation exports are joined on this identifier, so all models see exactly the same windows. Windows with too little usable gaze are discarded before embedding (at least 30 % valid frames and at least 32 valid frames), and remaining gaps inside a retained window are linearly interpolated. Of roughly 7.3 million recorded windows, only those inside a labelled interval have an unambiguous target and are usable. Handcrafted features are available for 177,244 of them, 157,685 also pass the encoder quality control, and 140,531 carry a numeric engagement rating. The final modelling cohort accounts for about 2 % of all recorded windows and contains between 2,136 and 25,447 windows per participant.

Display resolutions range from 1920×1080 to 3840×2160 pixels. Resolution differs between participants, and one participant even used two different resolutions across recordings, thus gaze coordinates are normalised to $[0, 1]$ using each source recording’s declared resolution. Out-of-screen coordinates are retained for modelling. Overall, 94.7 % of samples fall on the display, and only those enter the descriptive density map in Figure 1(a), which shows the expected central concentration together with broad coverage of the screen.

Four representations are computed from these windows. (1) **Raw** keeps the five channels, truncating longer windows to 120 samples and padding shorter ones with their final value before flattening them into a 600-dimensional vector. (2) **Handcrafted features** start from 199 exported columns. (here, we should start by a short simple explanation of how we created the handcrafted features and what groups we have. the current statement is just weird.) We remove 18 identifier, timestamp and questionnaire columns so that survey metadata cannot leak into the model, leaving 181 numeric candidates covering blink counts and durations, fixation and saccade statistics, pupil size and velocity moments, eye–screen distance statistics and per-channel missingness. (we don’t want to explain what we "remove". if we remove something we didn’t even need in the first place we don’t even want to mention we created it and then removed it. if we removed something like an ID, we should simply mention we didn’t use that in the training) (3) **GazeMAE embeddings** are computed with the public micro–macro autoencoder checkpoints applied to gaze position and velocity resampled to 500 Hz, with the 128-dimensional position and velocity codes concatenated into 256 dimensions [1]. (4) **MOMENT embeddings** are calculated with the frozen MOMENT-1-LARGE encoder applied to the same five channels with an input length of 512 and mean pooling, giving 1,024 dimensions [4]. Both encoders are used out of the box with no weight updates.

3.2 Experimental Design

Engagement ratings are strongly person-specific, with participant means spanning 2.2 to 4.9 on the 0–6 scale (Figure 1(b)). An absolute threshold would therefore mostly encode participant identity. We binarise each participant’s ratings relative to their own level. Inside every fold, the ratings of each training participant are mean-centred separately, and the held-out participant’s ratings are centred with the unweighted mean of the training participants’ means. A window is positive when its centred rating exceeds zero. Over the cohort this gives 56.3 % positive windows, ranging from 33 % to 72 % per participant.

The primary protocol is leave-one-subject-out cross-validation over the 17 participants. Everything data-dependent is fitted inside the fold. Handcrafted-feature filtering, median imputation and standardisation are estimated on the training participants and then applied with unchanged parameters to the held-out participant. The filtering drops columns with more than 60 % missing values, zero variance, or absolute Pearson correlation above 0.8 with a retained column, leaving 58–60 of the 181 feature candidates.

Three predictors are run on every representation: a most-frequent baseline, a random forest (300 trees, balanced class weights) and an MLP (1028–512–256–128–64 units, GELU activations, layer normalisation, Adam with learning rate 2×10^{-4} and weight decay 10^{-3}, batch size 1024, at most 200 epochs with

Table 1: Leave-one-subject-out results for binary engagement inference, averaged over the 17 held-out participants. SD is the standard deviation across participants and Δ is the improvement in accuracy over the majority baseline, paired within folds. Balanced accuracy is relative to 0.500 by construction. Accuracy uses all 17 folds. The remaining metrics use the 16 folds whose test participant is not single-class. M-F1 is macro-F1. Best value per column in bold.

	Acc.	SD	Δ	Bal. acc.	M-F1	AUC
Majority baseline	.524	.346	—	.500	.326	.500
Raw signals (600-d)						
Random forest	.527	.147	+.003	.529	.460	.546
MLP	**.538**	.113	**+.014**	**.536**	**.468**	**.551**
Handcrafted features (58–60-d)						
Random forest	.528	.230	+.005	.508	.418	.527
MLP	.531	.145	+.008	.521	.451	.535
GazeMAE embeddings (256-d)						
Random forest	.535	.155	+.012	.509	.441	.513
MLP	.520	.107	−.004	.493	.435	.494
MOMENT embeddings (1,024-d)						
Random forest	.521	.286	−.002	.501	.380	.504
MLP	.507	.274	−.017	.497	.362	.504

early stopping after 15 epochs without improvement). (use consistent terminology. replace most-frequent baseline with majority baseline here. also check other terminology for consistency. if you are unsure for some word, ask me.) A single seed is used throughout. The representation × model grid checks whether the representation ranking is robust across models.

We report accuracy as the primary metric, together with its improvement over the baseline paired within folds, and balanced accuracy, macro-F1 and ROC-AUC as secondary discrimination measures. All metrics are aggregated as unweighted means over participants. One participant self-reported every window below the training centre, so that test fold contains a single class. Therefore, accuracy is defined for all 17 folds and the remaining metrics for 16. As a separate within-participant check, we also run a one-step persistence predictor within recordings, carrying the last observed label forward in rolling blocks of 300 calibration and 90 test windows.

4 Results

Table 1 reports the main comparison. Every representation–model combination sits close to the majority baseline. The best result is the raw signal with the MLP, at 0.538 accuracy, 0.014 above the paired baseline, and it is also best on balanced accuracy (0.536), macro-F1 (0.468) and ROC-AUC (0.551). Handcrafted features come next and improve on the baseline with both classifiers (up to 0.531 accuracy and 0.535 AUC). GazeMAE reaches 0.535 accuracy with the random forest and falls to 0.520, below the baseline, with the MLP. MOMENT is at or below the baseline on every metric, and its AUC (0.504) is indistinguishable from chance.

Across all four metrics, the ordering is raw ≥ handcrafted > GazeMAE > MOMENT, but the absolute accuracy differences of 0.003 to 0.017 are much smaller than the spread across held-out participants, whose accuracy standard deviations range from 0.11 to 0.29. The MLP improves raw and handcrafted inputs but lowers both embedding results. A flexible head may help when a representation already carries useful signal. The effect is small, so this interpretation remains tentative.

Figure 2 gives a geometric view of the same difficulty. In two-dimensional t-SNE projections, none of the four representations arranges windows by engagement rating. The colours are mixed everywhere, in the learned embeddings as much as in the raw signal. Colouring the identical projections by participant reveals some local grouping for raw signals and GazeMAE. This suggests that part of the structure captured by these representations is person-specific.

As a check on the label series, a one-step persistence predictor applied within recordings reaches 0.999 accuracy, 0.960 balanced accuracy and 0.965 macro-F1 over 1,434 temporal blocks. Each questionnaire answer is broadcast unchanged to every window of the preceding 15-minute interval (Section 3), so most consecutive window pairs share a label by construction. Because the predictor uses observed labels within participants, its result cannot be compared directly with the unseen-participant results above.

5 Discussion

On this data, the raw multichannel window is the most informative of the four representations for binary engagement inference in unseen office workers, with handcrafted features close behind and no improvement from the two frozen embeddings. The small margins over the majority baseline show that three seconds of gaze does not make momentary self-reported engagement well predictable across people, whichever representation is used. Representation differences are modest but more consistent than classifier effects. (weird sentence, make it easier and clearer.)

The failure of the frozen encoders to improve on simpler inputs is best read as a mismatch between pretraining and task. (i don't like the wording "best read". change to simpler language. or maybe "interpreted" or "we interpret".) GazeMAE was pretrained on controlled stimulus-viewing corpora and MOMENT on generic time-series benchmarks. Neither has seen day-long, spontaneous office gaze with its dropouts, idle stretches and heterogeneous displays. These out-of-the-box results concern frozen transfer in this setting and leave adapted representation learning open. Domain adaptation or fine-tuning on in-domain gaze is the natural next step, especially for GazeMAE given its classifier-dependent behaviour (Section 4). (make the sentence simpler, more readable.)

For practice, raw signals are the empirical winner and the cheapest option: no feature engineering, no pretrained checkpoint, no GPU inference pass, and 600 dimensions against MOMENT's 1,024. Handcrafted features should still be considered. They finished a close second, they remain interpretable, and our earlier feature-based analyses found them effective on related eye-tracking targets [3, 2], so they are still worth trying on a similar problem. The results give little reason to choose a frozen general-purpose encoder by default. (in general make the paper less "artisticly written" regarding the wording. mostly it is fine, but this sentence for example "give little reason to" and some other similar cases make it look like i want to write a book. but no. we are writing a SCI paper, not a fiction book.)

Several limitations bound these conclusions. The study uses one unpublished and still-growing dataset with 17 participants. The target is a noisy self-report assigned retrospectively to a 15-minute interval and binarised by within-participant centring. The results depend on one windowing scheme, one seed, one set of quality-control rules and two specific public checkpoints.

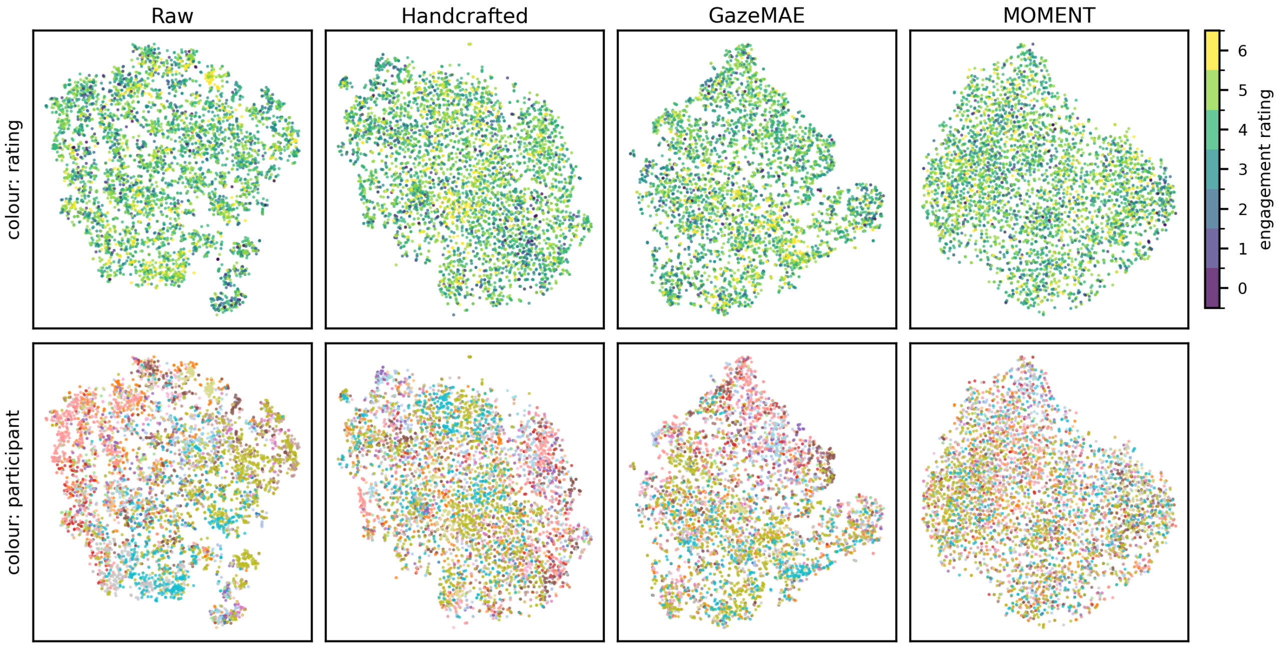

Figure 2: t-SNE projections of the four representations over a common sample of 5,000 windows, coloured by engagement rating (top) and by participant (bottom). The projections show exploratory geometry and should not be interpreted as evidence of separability under a trained classifier. (the last sentence is too much general. make a better claim, different. such as: we can observe by eye that the data is quite subject-specific based on the raw t-sne plot colored per subject. implying difficult data and justifying to some extent that the raw representation works best because other representations seem to smooth out the noise. comment: we do not need to say all these ideas in the figure caption. you can put some to the main text in appropriate place. of course, make the wording better.)

The density map and the projections are descriptive only. Future work should validate the comparison on further datasets, adapt or fine-tune the encoders on in-domain gaze, explore calibrated personalisation for new users, and test longer windows and aggregated targets that match the granularity at which engagement is actually reported.

6 Conclusion

We compared four eye-tracking representations under one identical, leakage-safe leave-one-subject-out protocol for momentary engagement inference in office workers: raw multichannel windows, handcrafted ocular features, frozen GazeMAE embeddings and frozen MOMENT embeddings. Raw windows were the simplest and cheapest representation, and they achieved the best result with 0.538 accuracy and 0.551 ROC-AUC on unseen participants. The frozen encoders did not repay their additional cost out of the box. (again seems like we are writing a book) For practitioners working with limited labelled data, the evidence favours raw windows or handcrafted features as a starting point and adapting frozen foundation-model embeddings before use.

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